# Yiwei Liu

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## Education

**ELLIS PhD Program** Stuttgart / Munich, Germany
*Jointly Supervised PhD Student (IMPRS-IS, University of Stuttgart)* *Incoming*

- **Supervisors:** Prof. Dr. Andreas Bulling (Collaborative Artificial Intelligence Group) and Dr. Eric Schulz (Institute for Human-Centered AI)

**École Polytechnique Fédérale de Lausanne (EPFL)** Lausanne, Switzerland
*MSc in Digital Humanities (GPA: 5.35/6)* *Sept 2023 – Oct 2026*

**Beijing Institute of Technology (BIT)** Beijing, China
*BSc in Statistics (GPA: 88/100, Major GPA: 90/100)* *Sept 2019 – June 2023*

## Publications And Technical Reports

- The Illusion of Efficacy: Diagnosing the Generative Side Effects of Knowledge Editing
  *Yiwei Liu*, Thibault Laugel, Tim Arni, and Marcin Detyniecki
  *Grounding Language Models (GroundLM) Workshop, EMNLP 2026.*
- TactfulToM: Do LLMs Have the Theory of Mind Ability to Understand White Lies?
  *Yiwei Liu*, Emma Jane Pretty, Jiahao Huang, Saku Sugawara
  *EMNLP 2025, Main Conference.*
- Business Model Construction Logic of Research-based Platforms *(Chapter 12)*
  Co-author, contributed to data analysis; conducted under Prof. Wei Wei at Peking University.
  *China Machine Press, 2024 (ISBN: 978-7-111-73714-8)*

## Research Experience

**Workplace GenAI Use: Human-AI Collaboration through the Lens of Cognitive Delegation** *Fen 2026 – Aug 2026*
*Master thesis | Supervisors: Dr. Thibault Laugel, Dr. Xiaofu Jin, and Prof. Frédéric Kaplan | AXA GO Research*

- Analyzed 18,000 messages from 4,461 conversations with an internal GenAI assistant at a leading global financial-services company to characterize the cognitive work employees delegate to LLMs and how much task framing they externalize in prompts, supporting organization-level monitoring of workplace cognitive delegation patterns.
- Modeled task- and employee-level usage patterns using mixed-effects regression on HR-linked data; currently preparing a user study to identify pain points in workplace LLM use and inform product and interaction design.

**Knowledge Editing and Generative Reliability of LLMs** *Aug 2025 – Feb 2026*
*Supervisor: Dr. Thibault Laugel | AXA GO Research, Switzerland*

- Evaluated five knowledge-editing methods (including *ROME*, *MEMIT*, and *GRACE*) across two LLMs using free-form generation and multi-turn dialogue; combined rule-based checks and *LLM-as-a-Judge* to classify failures overlooked by standard edit-success metrics, including repetition, factual contradictions, and hallucinations.
- Probed layer-wise representation changes using *LogitLens* and activation-space geometry analyses to investigate potential mechanisms underlying knowledge-editing failures.

## Internship Experience

**Beijing Xianfeng Changqing Venture Capital (K2VC)** Beijing, China
*Investment Research Intern | Supervisor: Runxin Yang* *Dec 2022 – June 2023*
∗ Conducted evaluations of AI and robotics startups, tracking paradigm shifts to inform investment decisions.

**MiraclePlus (Former Y Combinator China)** Beijing, China
*Investment Research Intern |Supervisor: Jack Jin, Dr. Qi Lu* *Dec 2021 – Feb 2022*
∗ Engaged with leading AI labs in China to track frontier developments in AI; performed technical evaluation on investment targets including TensorOpera AI and Thewake Systems, assessing research-to-product pathways.

**JD.com Explore Academy (AI Lab)** Beijing, China
*Algorithm Engineer Intern | Supervisor: Dr. Chang Li* *June 2021 – Aug 2021*

* Developed a Proof-of-Concept for JD.com × Nvidia Omniverse Virtual Retailer that collects eye-tracking and interaction data and applies multimodal deep learning methods for behavior prediction to optimize shelf layout; contributed to internal AI research strategy, focusing on multimodal and trustworthy AI for e-commerce.

## Selected Course Projects

**Developed a Specialized LLM Tutor for EPFL Course** *Feb 2024 – May 2024*

*Modern Natural Language Processing (CS-552) | Instructor: Prof. Antoine Bosselut*

* Fine-tuned Phi-2 with DoRA (PEFT) and Direct Preference Optimization (DPO) for efficient alignment; integrated Retrieval-Augmented Generation (RAG) for context-aware responses and applied quantization for deployment.

**Advantage Actor-Critic (A2C) for the CartPole Problem** *Feb 2024 – May 2024*

*Course Project for Reinforcement Learning (CS-456) | Instructor: Prof. Gerstner Wulfram*

* Implemented the Advantage Actor-Critic (A2C) algorithm on the CartPole environment, leveraging parallel processing and multi-step returns to improve training efficiency.

**Extending Text2Image Model to Multimodal Model** *Sept 2023 – Dec 2023*

*Foundation of Digital Humanities (DH-405) | Instructor: Prof. Frédéric Kaplan*

* Integrated ImageBind and Stable Diffusion into a multimodal pipeline; developed a benchmark to evaluate cultural understanding and image–text coherence, revealing limited generalization in cultural concept understanding.

## Skills

- **Programming & Frameworks:** Python, PyTorch, TensorFlow, Sklearn, Pandas, NumPy, JavaScript/HTML, Git, Linux.
- **Languages:** English (Proficient), Chinese (Native), French (Beginner), German (Beginner).